Unit 1

Lesson Two

Higher derivatives of the function

If $y = f(x)$ is a differentiable function with respect to x for a number of times.

Then we can get the first derivative and the second derivative , ... and so on as the following :

* The first derivative of the function is $\frac{dy}{dx}$ or $\dot{y}$ or $f'(x)$

* The second derivative of the function is : "the derivative of the first derivative".

  i.e. $\frac{d}{dx}\left(\frac{dy}{dx}\right)$

  It is denoted by one of the symbols $\frac{d^2y}{dx^2}$ or $\ddot{y}$ or $f''(x)$

* The third derivative of the function is "the derivative of the second derivative".

  i.e. $\frac{d}{dx}\left(\frac{d^2y}{dx^2}\right)$

  It is denoted by one of the symbols $\frac{d^3y}{dx^3}$ or $\dddot{y}$ or $f'''(x)$ ... etc.

For example :

If : $y = x^4 - 3x^3 + 5x^2 - 4x + 7$

Then : $\dot{y} = 4x^3 - 9x^2 + 10x - 4$, $\ddot{y} = 12x^2 - 18x + 10$

, $\dddot{y} = 24x - 18$, $\frac{d^4y}{dx^4} = 24$, $\frac{d^5y}{dx^5} = \text{zero}$

and any derivative after the 5th derivative $= 0$

Remarks

1. $\frac{d^2y}{dx^2}$ is read "d two y d x two"

2. The derivatives of a function starting by the second derivative is called higher derivatives and the derivative of order "n" is written as the following : $y^{(n)} = \frac{d^ny}{dx^n} = f^{(n)}(x)$ where n is positive integer.

3. Note the difference between $\frac{d^2y}{dx^2}$ and $\left(\frac{dy}{dx}\right)^2$ such that :

   * $\frac{d^2y}{dx^2}$ is the second derivative of y

   * $\left(\frac{dy}{dx}\right)^2$ is the square of the first derivative of y

4. $* \dfrac{dy}{dx} \times \dfrac{dx}{dy} = 1$ But $\dfrac{d^2y}{dx^2} \times \dfrac{d^2x}{dy^2} \neq 1$

$* \dfrac{dy}{dz} \times \dfrac{dz}{dx} = \dfrac{dy}{dx}$ But $\dfrac{d^2y}{dz^2} \times \dfrac{d^2z}{dx^2} \neq \dfrac{d^2y}{dx^2}$

"that is no chain rule of the second derivative".

5. $\dfrac{d}{dx}\left[y \dfrac{dy}{dx}\right]$ = 1st × derivative of the 2nd + 2nd × derivative of the 1st

$= y \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right) + \dfrac{dy}{dx}\dfrac{dy}{dx} = y \dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2$

6. The rate of change of the slope of the tangent to the curve

$y = f(x)$ equals $\dfrac{d}{dx}\left(\dfrac{dy}{dx}\right) = \dfrac{d^2y}{dx^2}$

Example 1

If $y = \frac{1}{4}x^4 + 2x^3 - x^2 + 3x + 1$, find : $\dfrac{d^2y}{dx^2}$ at $x = 2$

Solution

$\dfrac{dy}{dx} = x^3 + 6x^2 - 2x + 3$ $\quad \therefore \dfrac{d^2y}{dx^2} = 3x^2 + 12x - 2$ $\quad \therefore \left[\dfrac{d^2y}{dx^2}\right]_{x=2} = 34$

Example 2

Find the 3rd and the 4th derivative for each of the following :

1. $y = x^3 - \sin 2x$ **2.** $y = \dfrac{x}{x-1}$ where : $x \neq 1$

Solution

1. $\because y = x^3 - \sin 2x$ $\quad \therefore \dfrac{dy}{dx} = 3x^2 - 2\cos 2x$

$\therefore \dfrac{d^2y}{dx^2} = 6x + 4\sin 2x$ $\quad \therefore \dfrac{d^3y}{dx^3} = 6 + 8\cos 2x$, $\dfrac{d^4y}{dx^4} = -16\sin 2x$

2. $\because y = \dfrac{x}{x-1}$ $\quad \therefore \dfrac{dy}{dx} = \dfrac{(x-1)-x}{(x-1)^2} = \dfrac{-1}{(x-1)^2} = -(x-1)^{-2}$

$\therefore \dfrac{d^2y}{dx^2} = 2(x-1)^{-3}$ $\quad \therefore \dfrac{d^3y}{dx^3} = -6(x-1)^{-4} = \dfrac{-6}{(x-1)^4}$

$\therefore \dfrac{d^4y}{dx^4} = 24(x-1)^{-5} = \dfrac{24}{(x-1)^5}$

Example 3

If $y^2 = x^2 (4 - x)$, prove that : $y \frac{d^2 y}{d x^2} + \left(\frac{d y}{d x}\right)^2 + 3 x = 4$

Solution

$\because y^2 = 4 x^2 - x^3$ (by differentiating with respect to x)

$\therefore 2 y \frac{d y}{d x} = 8 x - 3 x^2$ (dividing by 2)

$\therefore y \frac{d y}{d x} = 4 x - \frac{3}{2} x^2$ (by differentiating with respect to x)

$\therefore y \frac{d^2 y}{d x^2} + \left(\frac{d y}{d x}\right)^2 = 4 - 3 x$ $\quad\quad$ $\therefore y \frac{d^2 y}{d x^2} + \left(\frac{d y}{d x}\right)^2 + 3 x = 4$

Example 4

If $y = x^2 + 7 x - 1$, $z = x^2 - 3 x + 15$, find : $\frac{d^2 y}{d z^2}$ at $x = 2$

Solution

$\because y = x^2 + 7 x - 1$ $\quad\quad$ $\therefore \frac{d y}{d x} = 2 x + 7$

, $\because z = x^2 - 3 x + 15$ $\quad\quad$ $\therefore \frac{d z}{d x} = 2 x - 3$

$\therefore \frac{d y}{d z} = \frac{d y}{d x} \div \frac{d z}{d x}$, $\quad\quad$ $\therefore \frac{d y}{d z} = \frac{2 x + 7}{2 x - 3}$ (by differentiating with respect to z)

$\therefore \frac{d}{d z}\left(\frac{d y}{d z}\right) = \frac{d}{d z}\left(\frac{2 x + 7}{2 x - 3}\right)$

$\therefore \frac{d^2 y}{d z^2} = \frac{d}{d x}\left(\frac{2 x + 7}{2 x - 3}\right) \times \frac{d x}{d z} = \frac{2 (2 x - 3) - 2 (2 x + 7)}{(2 x - 3)^2} \times \frac{1}{2 x - 3} = \frac{- 20}{(2 x - 3)^3}$

$\therefore \left(\frac{d^2 y}{d z^2}\right)_{x = 2} = \frac{- 20}{(2 \times 2 - 3)^3} = - 20$

Example 5

If $y = \sin x + \cos x$, prove that : $\frac{d^4 y}{d x^4} = y$

Solution

$\because y = \sin x + \cos x$

$\therefore \dot{y} = \cos x - \sin x$ $\quad\quad$ $\therefore \ddot{y} = - \sin x - \cos x$

$\therefore \frac{d^3 y}{d x^3} = - \cos x + \sin x$, $\frac{d^4 y}{d x^4} = \sin x + \cos x = y$

Example 6

If $x^2 + y^2 = 1$, prove that : $y^3 \frac{d^2y}{dx^2} + 1 = 0$

Solution

$\because x^2 + y^2 = 1$ (by differentiating with respect to x)

$\therefore 2x + 2y \frac{dy}{dx} = 0$ (divided by 2)

$\therefore x + y \frac{dy}{dx} = 0$ (1) $\therefore \boxed{\frac{dy}{dx} = \frac{-x}{y}}$

(by differentiating (1) with respect to x)

$\therefore 1 + y \frac{d^2y}{dx^2} + \frac{dy}{dx} \times \frac{dy}{dx} = 0$ $\therefore 1 + y \frac{x^2}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$

by substituting of $\frac{dy}{dx} = \frac{-x}{y}$

$\therefore 1 + y \frac{d^2y}{dx^2} + \left(\frac{-x}{y}\right)^2 = 0$ $\therefore 1 + y \frac{d^2y}{dx^2} + \frac{x^2}{y^2} = 0$

Multiplying by (y^2) : $\therefore y^2 + y^3 \frac{d^2y}{dx^2} + x^2 = 0$

, $\because x^2 + y^2 = 1$ $\therefore y^3 \frac{d^2y}{dx^2} + 1 = 0$

Example 7

If $xy = 2$, prove that : $x^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + y = 0$

Solution

$\because xy = 2$ $\therefore y = \frac{2}{x} = 2x^{-1}$

$\therefore \frac{dy}{dx} = -2x^{-2}$ $\therefore \frac{d^2y}{dx^2} = 4x^{-3}$

$\therefore x^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + y$

$= x^2 \times 4x^{-3} + 3x \times -2x^{-2} + 2x^{-1}$

$= 4x^{-1} - 6x^{-1} + 2x^{-1} =$ zero

Another solution :

$\because xy = 2$

$\therefore x\dot{y} + y = 0$ (multiplying by x)

$\therefore x^2\dot{y} + xy = 0$

$\therefore x^2\ddot{y} + 2x\dot{y} + x\dot{y} + y = 0$

$\therefore x^2\ddot{y} + 3x\dot{y} + y = 0$

Example 8

If $y = \sin a x$, discover the pattern of consecutive derivative , then find $y^{(37)}$, $y^{(2024)}$

Solution

$\because y = \sin a x$ $\quad\therefore \dot{y} = a \cos a x = a \sin\left(a x + \frac{\pi}{2}\right)$

, $\ddot{y} = -a^2 \sin a x = a^2 \sin(a x + \pi)$

, $y^{(3)} = -a^3 \cos a x = a^3 \sin\left(a x + \frac{3\pi}{2}\right)$

, $y^{(4)} = a^4 \sin a x = a^4 \sin(a x + 2\pi)$

$\therefore$ $\boxed{y^{(n)} = a^n \sin\left(a x + \frac{n\pi}{2}\right)}$

$\therefore y^{(37)} = a^{37} \sin\left(a x + \frac{37\pi}{2}\right) = a^{37} \sin\left(a x + \frac{\pi}{2}\right) = a^{37} \cos a x$

, $y^{(2024)} = a^{2024} \sin\left(a x + \frac{2024\pi}{2}\right) = a^{2024} \sin a x$

From the previous example we notice that :

1. If $y = \sin a x$, then $y^{(n)} = a^n \sin\left(a x + \frac{n\pi}{2}\right)$
2. If $y = \cos a x$, then $y^{(n)} = a^n \cos\left(a x + \frac{n\pi}{2}\right)$
3. If $\boxed{y = \sin a x}$ or $\boxed{y = \cos a x}$

   , then $\boxed{y^{(n)} = a^n y}$ where n is divisible by 4

Remark

If $y = x^n$, then :

1. $\frac{d^m y}{d x^m} = {}^nP_m \, x^{n-m}$ for every $n , m \in \mathbb{Z}^+ , n \geq m$
2. $\frac{d^m y}{d x^m} =$ zero for every $n , m \in \mathbb{Z}^+ , n < m$

For example : If $y = x^6$, then :

* $\frac{d^2 y}{d x^2} = {}^6P_2 \, x^{6-2} = 30 x^4$
* $\frac{d^6 y}{d x^6} = {}^6P_6 \, x^{6-6} = \lfloor 6 \, x^0 = \lfloor 6$
* $\frac{d^8 y}{d x^8} =$ zero

Algebraic pre-requirements for the natural exponential function and the natural logarithmic function

The Nabier's number (e)

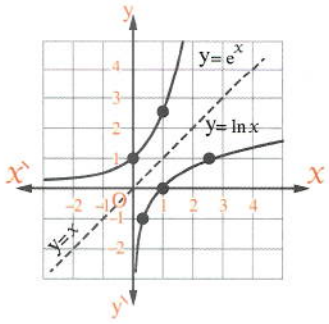
The Nabier's constant e is an irrational number , $2 < e < 3$,

$e = 1 + \frac{1}{\lfloor 1} + \frac{1}{\lfloor 2} + \frac{1}{\lfloor 3} + \cdots = \sum_{n=0}^{\infty} \frac{1}{\lfloor n}$ (Taylor's series)

$e = 2.71828$

⊙ The natural exponential function :

$f : \mathbb{R} \longrightarrow \mathbb{R}^+$ where $f(x) = e^x$



is exponential function whose base is e and it is one - to - one its domain = $\mathbb{R}$, its range = $\mathbb{R}^+$, its curve passes through $(0 , 1)$, $(1 , e)$

⊙ The natural logarithmic function :

$f : \mathbb{R}^+ \longrightarrow \mathbb{R}$ where $f(x) = \ln x$

is logarithmic function base "e" ($\log_e x = \ln x$) and it is one - to - one function its domain $\mathbb{R}^+$, its range = $\mathbb{R}$ its curve passes through $(1 , 0)$, $(e , 1)$

Remarks

1. If $x \in \mathbb{R}^+$, $y \in \mathbb{R}$, then $y = \ln x \Leftrightarrow x = e^y$
2. The natural exponential function is inverse function of natural exponential function.
3. From the previous we get : $\lim_{x \to \infty} \ln x = \infty$, $\lim_{x \to 0^+} \ln x = -\infty$

Some properties of natural logarithm :

If $x , y \in \mathbb{R}^+$, $n \in \mathbb{R}$ under condition the base $\in \mathbb{R}^+ - \{1\}$ then :

1. $\ln e = 1$
2. $\ln 1 =$ zero
3. $\ln x^n = n \ln x$
4. $e^{\ln x} = x$
5. $\ln x y = \ln x + \ln y$
6. $\ln \frac{x}{y} = \ln x - \ln y$
7. $\log_y x = \frac{\ln x}{\ln y}$
8. $\ln x \times \log_x e = 1$

Example

Solve the following equations :

1. $\ln x + \ln (x+2) = \ln (x+6)$

2. $e^{3x-4} = 23$ **3.** $2e^{x} + 2e^{-x} = 5$

Solution

1. $\because \ln x + \ln (x+2) = \ln (x+6)$

$\therefore \ln x (x+2) = \ln (x+6)$ $\therefore x(x+2) = x+6$

$\therefore x^2 + x - 6 = 0$, $\therefore (x-2)(x+3) = 0$

$\therefore x = 2$ (satisfy) or $x = -3$ (refused)

2. $\because e^{3x-4} = 23$ by taking ln for each side

$\therefore \ln e^{3x-4} = \ln 23$

$\therefore (3x-4) \ln e = \ln 23$

$\therefore 3x - 4 = \ln 23$

$\therefore 3x = 4 + \ln 23$

$\therefore x \simeq 2.38$

Remark

To write $4 + \ln 23$ on the calculator :



3. $\because 2e^{x} + 2e^{-x} = 5$ Multiplying by e^{x}

$\therefore 2e^{2x} - 5e^{x} + 2 = 0$

$\therefore (2e^{x} - 1)(e^{x} - 2) = 0$

$\therefore e^{x} = \frac{1}{2}$ or $e^{x} = 2$

$\therefore x = \ln \frac{1}{2} \simeq -0.69$ or $x = \ln 2 \simeq 0.69$